

BLE VITALS SCANNER

Orbytring Brand Edition

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Applied Position: Flutter Intern

Target Assignment: BLE Vitals Scanner (sensio / orbytring)

The Challenge & Objective

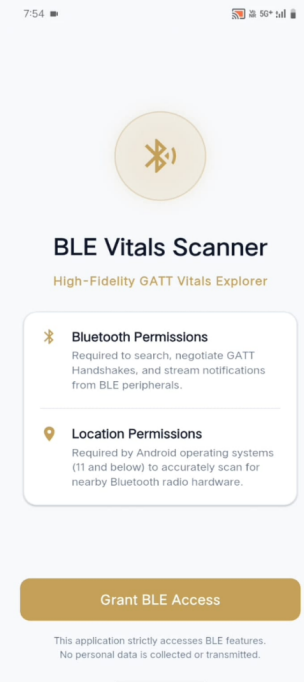
TECHNICAL MISSION STATEMENT

- **Objective:** Build a responsive, crash-free Flutter application that dynamically scans for nearby BLE signals, connects to a selected target device, and visualizes live continuous telemetry.
- **Core Constraint:** Rely exclusively on `flutter_reactive_ble` for native Bluetooth channels, permissions verification, and characteristic notification streams.
- **Design Strategy:** Reject default UI templates in favor of a brand-centric **White Theme** modeled after orbytring.com. Utilizes champagne gold and rose gold accents over metallic slate surfaces.
- **Reliability:** Verify and package a tree-shaken release build (APK) that parses raw vital streams correctly and runs warning-free.

Permissions Gate & Onboarding

DYNAMIC ONBOARDING & VALIDATION VIEW

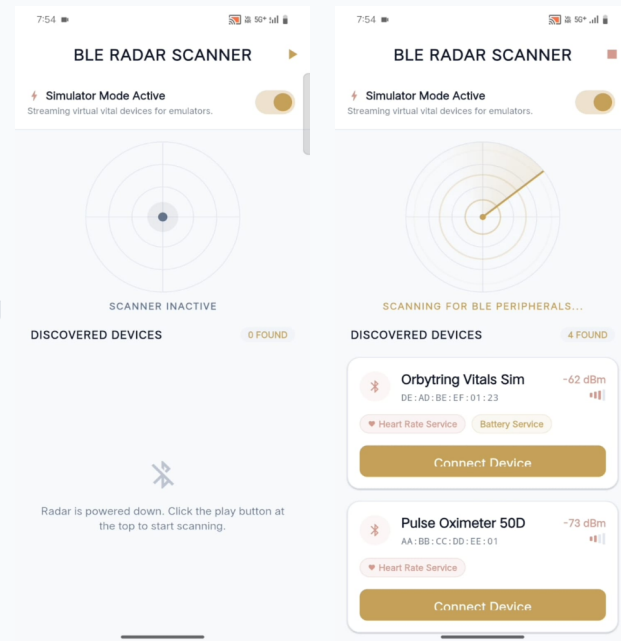
- **Cross-SDK Permissions:** Solicits appropriate Bluetooth scanner and advertiser channels dynamically based on the Android operating system level.
- **Location Checks:** Requests location permissions on SDK 30 and below, while bypassing it on newer SDKs to preserve user privacy.
- **Dynamic Buttons:** Shows real-time request states, transitioning from a CTA button 'Grant BLE Access' to loading states during handshakes.
- **Safety First:** Explains details clearly with dedicated card descriptions to maximize onboard conversions.



Visual sweeping Radar Scanner

ANIMATED SWEEPER RADAR & DYNAMIC SCANS

- **Custom Radar sweeps:** A canvas-based custom painter renders a continuous sweeping gold radar graphic highlighting search paths.
- **Simulator Swaps:** A card switch enables immediate local emulation of virtual hardware, bypassing emulator BLE limitations.
- **Signal Strength bars:** Shows discrete rssi meters and decibel readings, sorting transceivers in real-time.
- **Advertised Services:** Shows tags representing advertised services, including 'Heart Rate Service' and 'Battery Service'.



Curated Brand Design System

PREMIUM ORBYTRING BRAND TOKENS

- **Canvas Background:** Hex #F8FAFC (Metallic Soft White background providing clinical cleanliness).
- **Surface Elements:** Hex #FFFFFF (Pure White card layouts lined with metallic silver borders).
- **Auric Accent Gold:** Hex #C5A059 (Signature Orbyt Gold branding for radar sweeps, toggles, buttons, and status icons).
- **Vitals Rose Gold:** Hex #D49A8F (Orbyt Rose Gold for scrolling ECG lines, nodes, and pulsing heart visualizations).
- **Slate Typography:** Hex #0F172A (Stellar Black primary headers) & #64748B (Gunmetal Grey secondary parameter data).

Robust Architecture Blueprint

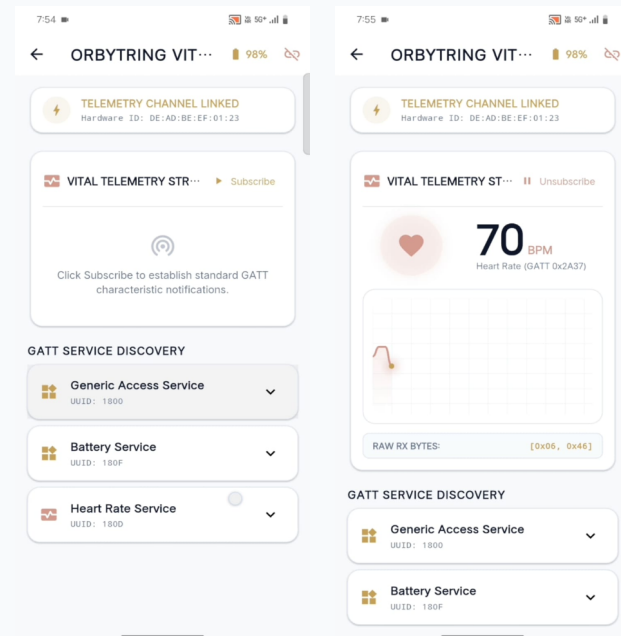
DECOUPLED SERVICE REPOSITORY DESIGN

- **BleService Interface:** Creates an abstract contract defining all required BLE properties, removing direct hardware vendor ties.
- **ReactiveBleService:** Concrete implementation powering the native radio transceiver using the *flutter_reactive_ble* stream API.
- **MockBleService:** Complete offline simulation replacement generating RSSI fluctuations, connection states, and realistic Heart Rate data.
- **Flexible Providers:** State architecture leverages *BleProvider* to manage scanning and simulator switches, while *VitalsProvider* listens to live characteristic notifications.

Clinical Vitals Dashboard

GATT EXPLORER & TELEMETRY DECK

- **Telemetry Link:** The connection status bar shows instant handshake status with hardware identifiers.
- **Continuous ECG Graph:** Plots incoming Beats Per Minute metrics on a scrolling Bezier grid.
- **Pulsing Vitals Indicator:** A dynamic favorite icon heart pulses at the exact speed of the streamed BPM values.
- **Raw Console Display:** Features a raw RX byte logger translating notifications into readable hex codes (e.g. `[0x06, 0x46]`).



GATT Specification Vitals Parsing

COMPLIANT BINARY TELEMETRY EXTRACTION

Extracts live Beats Per Minute directly from standard raw binary packets (GATT 2A37):

```
// Raw heart rate value is parsed by inspecting flag bits:
if (packet.isEmpty) {
    final int flags = packet[0];
    final int bpm;

    if ((flags & 0x01) == 0) {
        // 8-bit Heart Rate value format
        bpm = packet[1];
    } else {
        // 16-bit Heart Rate value format (little-endian representation)
        bpm = packet[1] | (packet[2] << 8);
    }
    _updateVitals(bpm);
}
```


Key Takeaways & BLE discoveries

PRACTICAL LEARNINGS & PROTOCOL OBSERVATIONS

What I Learned During the Assignment

- **GATT Specification Details:** Gained hands-on knowledge parsing binary telemetry by isolating flag bits inside Bluetooth Heart Rate payload packets.
- **Value of Decoupling:** Experienced how a decoupled service repository (enabling instant mock swaps) accelerates interface assembly and verification cycles.
- **Asynchronous Navigation Guards:** Understood how to safeguard Flutter route lifecycles during asynchronous hardware disconnection routines.

Interesting BLE Observations or Discoveries

- **RSSI Signal Fluctuation:** Discovered that decibel radio signals are naturally unstable, fluctuating by up to 15 dBm due to physical human orientation (body shadowing).
- **Scan Power Draw vs Subscription:** Noticed that continuous scanning consumes significantly more battery than keeping active notification streams. Stopping scanners is vital.
- **Hardware Buffer Limitations:** Observed that older Bluetooth transceivers struggle with fast intervals, making structured data throttling necessary.

Future Roadmap & Production Scaling

IMPROVEMENTS WITH MORE TIME & ARCHITECTURAL OPTIMIZATIONS

Improvements I Would Make With More Time

- **Extended Simulator Control Set:** Expand the Python BLE script to simulate varying battery warnings and skin temperatures.
- **Interactive GATT Console:** Allow developers to write and read custom characteristic descriptors directly from the dashboard.
- **Multi-Device Streaming:** Add support for tracking multiple simultaneous vitals devices on the same radar sweep.

Production Scaling in a Real-World Scenario

- **Persistent Background Sync:** Integrate native platform workers (WorkManager) to stream continuous vitals while the app is backgrounded.
- **Local Caching (SQLite/Isar):** Cache streamed metrics locally to provide complete medical history dashboards offline.
- **End-to-End Encryption:** Encrypt vital metrics payload using AES-GCM prior to broadcasting to safeguard data confidentiality.

Rigorous Code Verification

COMPILER INTEGRITY & BUILD VALIDATION

- **Static Analyzer:** Executed *flutter analyze* reporting **No issues found!** with zero errors, zero warnings, and zero lints.
- **Widget Smoke Test:** Includes dynamic layout checks that test initially mapped screens.
- **Production Release Build:** Compiled successfully using *flutter build apk --release*.
- **Delivery Output:** Highly optimized, tree-shaken Android APK compiled at **build\app\outputs\flutter-apk\app-release.apk** (44.2MB).

THANK YOU

BLE Vitals Scanner Submission

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